

ROS2 perception graph

RobotCore perception is an optimised robotic perception stack built with RobotCore framework that leverages hardware acceleration to provide a speedup in your perception computations. API-compatible with ROS2 perception stack, RobotCore perception delivers high performance, real-time, and reliability to your robots' perception.

in number. Another problem with ASIC adaptation in robotics is that all of the subfields of robotics like localisation, perception, and control are currently the fields of active research, hence, it lacks a final architecture. Therefore, using ASICs becomes a problem because they have a very specific architecture that needs to be defined. They are also really expensive to develop, so currently it is not really worth the effort.

The Robot Operating System (ROS) is the de facto framework for robot application development. It is a toolset to build and manage robots through libraries for robotics debugging and visualisation utilities, orchestration tools, and communication piping. Nearly 55% of total commercial robots shipped in 2024 will have at least one ROS package installed, creating a large installed base of ROS-enabled robots.

ROS is inherently CPU-centric. There's an opportunity to build custom compute architectures for robots and make them faster with

hardware acceleration. Most ROS packages are subject to hardware acceleration and licensed for commercial use.

RobotCore framework: an open architecture for hardware acceleration in ROS2

RobotCore provides a development, build, and deployment experience for creating robot hardware and hardware acceleration kernels similar to the standard, non-accelerated ROS development flow. In the standard data flow of a ROS2 application, passing it through a RobotCore framework leaves you with two benefits. At the node level you get acceleration and you also get inter-process acceleration.

RobotCore framework targets three different verticals in the customer market. First being the development packages for robotics, second being package maintenance, and third is semiconductor manufactures. The robotics division of the semiconductor companies would want to follow the RobotCore framework as it has a huge community acceptance. Semiconductor companies would also benefit as they will be able to get their products to market quicker. Currently, FPGA GPUs are very hot in the robotics market and that is where silicon vendors could benefit tremendously with this framework.

One key thing that has been done to get more acceptance of hardware acceleration in the robotics community is to create

a hardware acceleration working group as well. This includes accelerator vendors such as AMD Xilinx, NVIDIA, and Analog devices.

From RobotCore framework to an open standard

All the silicon vendors like AMD, NVIDIA, and Microchip form the bottom layer as they enable users to use their chips. Above that lies the RobotCore framework where we basically create the build tools to quickly be accepted into the ROS layer. On top of this layer sits the UDP/IP layer, which essentially is the middleware framework for ROS itself. At the top sits the application itself (see Fig. 2). RobotCore framework enables the bottom two layers so that it becomes easy for roboticists to develop applications.

Benchmarking performance in ROS2

The purpose of open-sourcing the RobotCore framework is to enable competition in the market itself as silicon vendors would therefore develop chips that eventually help the roboticists to make better robots. We are also enabling this through a benchmarking effort. We perform tests on FPGA brought to us by silicon vendors themselves.

RobotCore framework is a new open architecture for hardware acceleration in ROS2 that promises to revolutionise the world of robotics. This innovative technology allows developers to build high-performance robots with faster and more efficient processing, paving the way for advanced robotics applications. With the RobotCore framework, the future of robotics is looking brighter than ever. **EFY**

This article is based on a tech talk at India Electronics Week 2022 by Gaurav Vikhe, Chief of Product, Acceleration Robotics. It has been transcribed and curated by Laveesh Kocher, a tech enthusiast at EFY with a knack for open source exploration and research

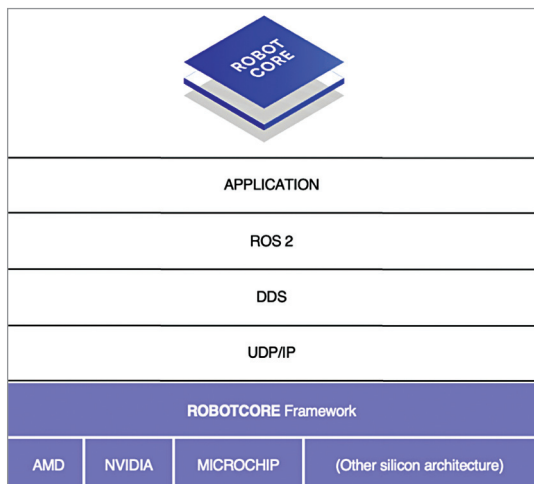


Fig. 2: ROS2 hardware acceleration architecture and convention